# SLEEP QUALITY PREDICTION USING LOGISTIC REGRESSION

# INTRODUCTION



GOOD | POOR

1. Sleep quality affects health, focus and productivity, and depends on daily habits.
2. This project predicts whether a person's sleep quality is Good or Poor using Machine Learning.
3. Inputs are age, gender, screen time, caffeine intake and exercise hours.
4. A Logistic Regression model classifies each person and gives a probability.
5. A Streamlit app turns the prediction into personalised advice.

# OBJECTIVE

**01**

Build a Logistic Regression model that predicts Good or Poor sleep quality.

**02**

Clean the data and explore it with histograms, bar charts and a heatmap.

**03**

Evaluate the model with accuracy, confusion matrix and ROC-AUC.

**04**

Deploy a Streamlit app with a sleep score gauge and personalised advice.

# TOOLS & TECHNOLOGY

**Python**

Main programming language

**pandas**

Loading and cleaning the data

**NumPy**

Numerical operations

**scikit-learn**

Logistic Regression, pipeline, metrics

**Matplotlib & Seaborn**

Histograms, heatmap, ROC curve

**Streamlit**

Interactive prediction web app

**joblib**

Saving the trained model

**CSV Dataset**

445 records, 7 columns

**Jupyter / VS Code**

Development and testing

# METHODOLOGY

1. **Dataset Collection –** Create a CSV dataset of 445 records with age, gender, screen time, caffeine, exercise and sleep quality.
2. **Data Cleaning –** Remove 25 duplicate rows, drop the id column and fill missing values (median for numbers, mode for gender).
3. **Exploratory Data Analysis –** Study patterns using histograms, bar charts, boxplots and a correlation heatmap.
4. **Encoding & Splitting –** Convert Good/Poor to 1/0, one-hot encode gender and split the data 80:20 into training and testing sets.
5. **Model Training –** Train a Logistic Regression model with feature scaling inside a scikit-learn pipeline.
6. **Model Evaluation –** Check performance using accuracy, confusion matrix, classification report and ROC curve.
7. **Streamlit Deployment –** Save the model and build a web app that predicts sleep quality and gives personalised advice.

# FUTURE ENHANCEMENT

1. Collect real data from surveys or wearable devices and add features like stress and sleep duration.
2. Compare Random Forest, Gradient Boosting and SVM with Logistic Regression.
3. Add user accounts to track habits and sleep score over time.
4. Deploy the app on the cloud with AI-generated personalised coaching.



# THANK YOU

Sweet dreams and better habits